1) NumPy Question Set (Complete Coverage)

1.1 Basics

a. Print sales.shape, sales.ndim, sales.dtype

b. Create zeros and ones array of shape (5,7)

1.2 Indexing & Slicing

a. Get sales of Shoes for all days

b. Get sales for Friday for all products

c. Get first 3 products and first 3 days

1.3 Aggregations

a. Total weekly sales per product (row sum)

b. Total sales per day (column sum)

c. Find max sales value and its index position

d. Mean sales per product

1.4 Reshape / Flatten

a. Flatten the matrix to 1D

b. Reshape into (7,5) and explain what it represents

1.5 np.where + conditions

a. Replace all values < 10 with 10

b. Find indices where sales > 35

1.6 Random + Simulation

a. Generate random noise matrix (5x7) and add it to sales

b. Generate random ratings between 1 and 5 for 10 customers

1.7 Linear Algebra (Basic)

a. Compute dot product of total sales per product with weights

b. Normalize sales using min-max scaling